Formal Specification: QQQ Dynamic Leverage Band Model

Derived directly from **QQQ_Dynamic_Leverage_Band_Model_v1.xlsx**

This specification describes the **same state transition already implemented in the workbook**, especially the dynamic-strategy columns **T:AG**. It is not a different strategy.

1. State variables

```
p_t = QQQ price on date t
h_t^- = shares held immediately before rebalancing
c_t^- = cash immediately before rebalancing
n_t^- = h_t^- * p_t = pre-rebalance position value
e_t^- = n_t^- + c_t^- = pre-rebalance equity
l_t^- = n_t^- / e_t^- = pre-rebalance leverage
```

Workbook mapping: **T** Shares_prev, **V** Cash_pre, **W** Equity_pre, **X** Leverage_pre, **Y** Rebalance_Action, **Z** Target_Pos, **AB** Trade_Shares, **AC** Shares_new, **AD** Cash_new, **AG** Leverage_new.

2. Fixed policy parameters

Target leverage L* = 3.0, lower threshold = 3.0, upper threshold = 3.3, and initial capital C_0 = 1,000,000. The workbook initializes a 3x long QQQ book with h_0 = 3C_0 / p_0 and c_0 = C_0 - h_0 p_0.

3. Trigger condition

```
If l_t^- < 3.0, action = BUY
If l_t^- > 3.3, action = SELL
Otherwise, action = HOLD
```

4. Rebalance rule

```
If action = HOLD: h_t = h_t^- and c_t = c_t^-
If action is BUY or SELL: target position value n_t* = 3 * e_t^-
Trade value Δn_t = n_t* - n_t^-
Trade shares Δh_t = Δn_t / p_t
Post-trade shares h_t = h_t^- + Δh_t
Post-trade cash c_t = c_t^- - Δn_t
```

5. Required postconditions

- **P1. Equity preservation:** after rebalance, e_t = e_t^-.
- **P2. Correct action sign:** BUY implies Δh_t > 0, SELL implies Δh_t < 0, HOLD implies Δh_t = 0.
- **P3. Target restoration:** under exact workbook arithmetic, any executed rebalance returns leverage to 3.0.
- **P4. No-op inside the band:** if 3.0 <= l_t^- <= 3.3, the economic state must remain unchanged.

6. Modeling assumptions

The workbook assumptions are preserved: no transaction costs, no slippage, no borrow interest, same-date observation/execution price, and fractional shares allowed.

7. Implementation note

A Solidity implementation uses fixed-point integers instead of real numbers. Therefore, the exact equality in P3 becomes an approximate claim: after rebalance, leverage is restored to the target *up to rounding error*, and the next action should be HOLD.